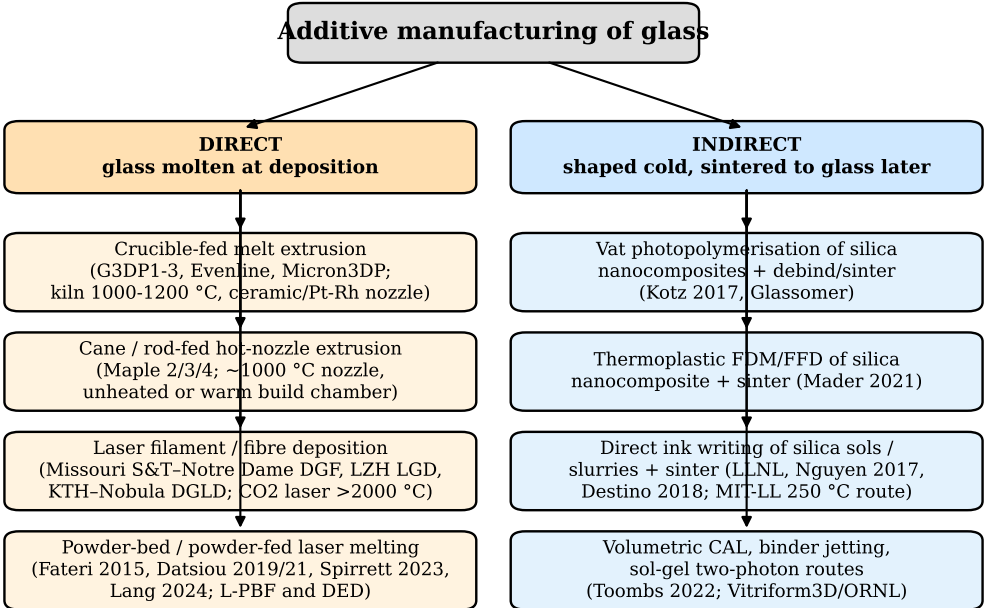
Additive manufacturing of glass
DIRECT
glass molten at deposition
INDIRECT
shaped cold, sintered to glass later
Crucible-fed melt extrusion
(G3DP1-3, Evenline, Micron3DP;
kiln 1000-1200 °C, ceramic/Pt-Rh nozzle)
Cane / rod-fed hot-nozzle extrusion
(Maple 2/3/4; ~1000 °C nozzle,
unheated or warm build chamber)
Laser filament / fibre deposition
(Missouri S&T–Notre Dame DGF, LZH LGD,
KTH–Nobula DGLD; CO2 laser >2000 °C)
Powder-bed / powder-fed laser melting
(Fateri 2015, Datsiou 2019/21, Spirrett 2023,
Lang 2024; L-PBF and DED)
Vat photopolymerisation of silica
nanocomposites + debind/sinter
(Kotz 2017, Glassomer)
Thermoplastic FDM/FFD of silica
nanocomposite + sinter (Mader 2021)
Direct ink writing of silica sols /
slurries + sinter (LLNL, Nguyen 2017,
Destino 2018; MIT-LL 250 °C route)
Volumetric CAL, binder jetting,
sol-gel two-photon routes
(Toombs 2022; Vitriform3D/ORNL)